

Experiment 14: Homography Matrix Transformation

AIM

To perform image transformation using the Homography Matrix.

PROCEDURE

1. Read the image.
2. Define four corresponding points.
3. Compute the homography matrix.
4. Warp the image.
5. Display the result.

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

rows, cols = img.shape[:2]

src = np.float32([[0,0],[cols,0],[0,rows],[cols,rows]])
dst = np.float32([[40,40],[cols-40,20],[20,rows-20],[cols-20,rows-40]])

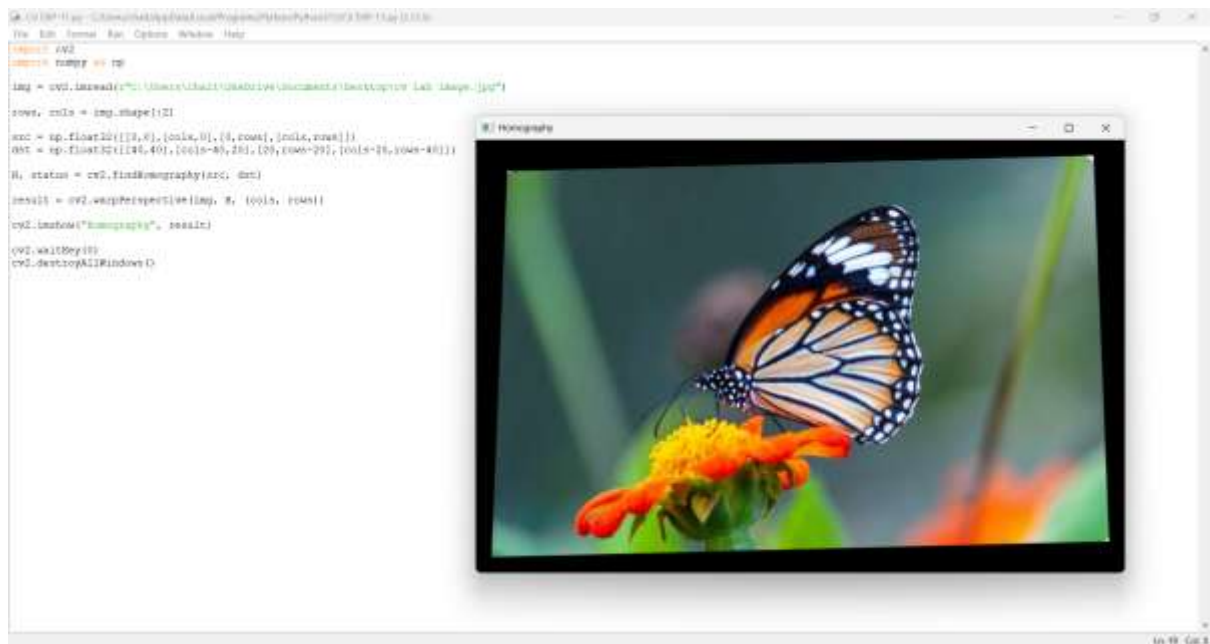
H, status = cv2.findHomography(src, dst)

result = cv2.warpPerspective(img, H, (cols, rows))

cv2.imshow("Homography", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT:



RESULT:

Perspective transformed Image is displayed.